Cloud Computing Tutorial - 4 and 5

CloudSim Simulation

Observation:

Tutorial - 4 and 5

What is cloudsim?

cloudsim is a simulation toolkit used to model and simulate cloud computing environments. Instead of building a real cloud infrastructure, you can use cloudsim to test and analyze cloud system virtually.

Why do we use cloudsim?

Cloudsim is used to

* Test resource allocation algorithms
* Evaluate task scheduling strategies
* Analyze performance.
* Simulate data centers, VMs and users
* Study load balancing techniques.

What can cloudsim simulate?

cloudsim models different parts of cloud system

* Data centers → physical infrastructure
* Hosts → servers inside data centers
* Virtual machines → Logical units running tasks
* cloudlets → Task/Jobs submitted by users
* Brokers → Manage task assignment to VMs.

Bafna Gold
Date: Page:

Key features:

* Supports large-scale cloud emulation
* Customizable policies
* Models network behaviour and energy consumption
* Supports visualization concepts

Advantages:
* Cost-effective
* Easy implementation
* Scalable
* Highly customizable
* Performance analyse

Prerequisites: JDK8 or higher
Eclipse IDE
cloudsim source code
Apache common math library

Add cloudsim source code to eclipse
manually copy and paste code into eclipse project
1. Open eclipse and create a new java project
2. paste cloudsim source code
3. refresh the project

Add required libraries
-add external jar files

Run a cloudsim example

Eg: cloudsim example.java in source folder output will display in console

Output

Starting cloudsimExample...
Initializing
Starting cloudsim version 3.0...
Datacenter_0 is Starting
:
Simulation completed
output status → success
cloudsim example 1 Finished!

CloudSim Simulation:

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>
```

Fig 1.1: JDK installation verification

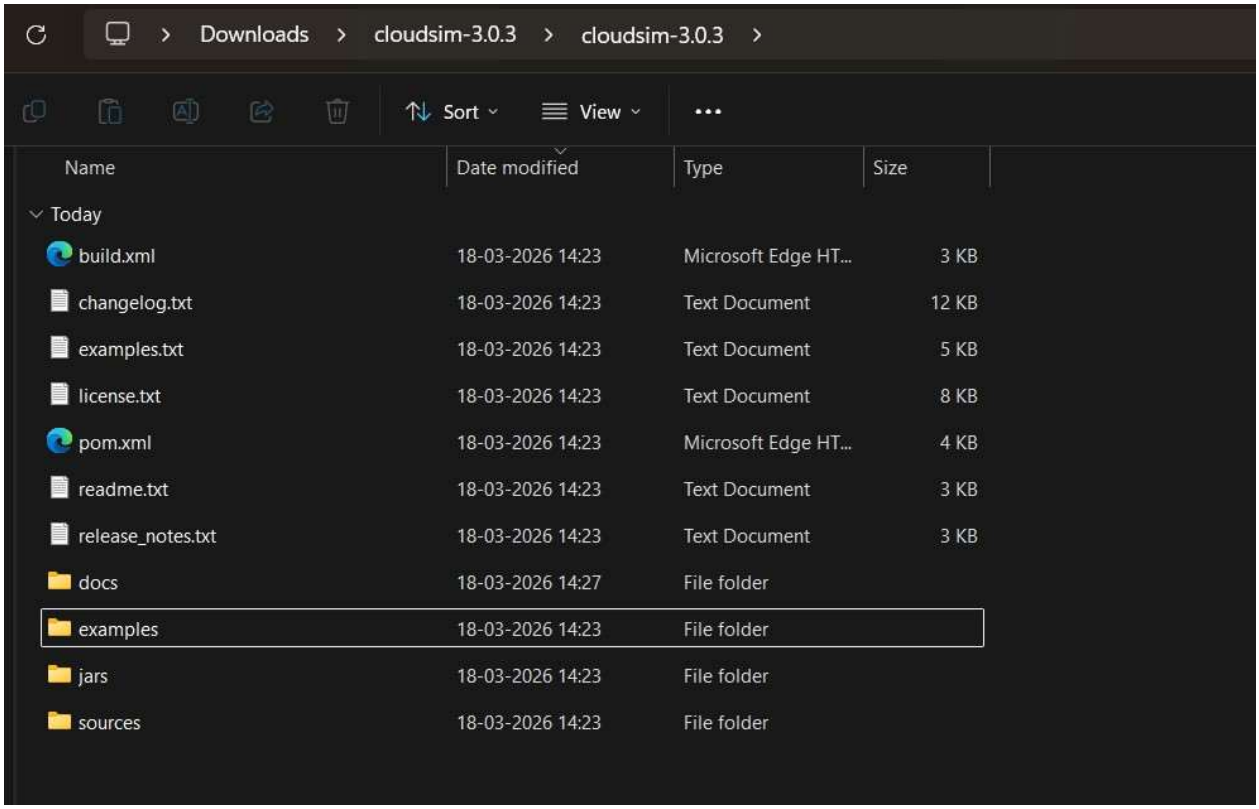

Fig 1.2: cloudsim zip folder extraction

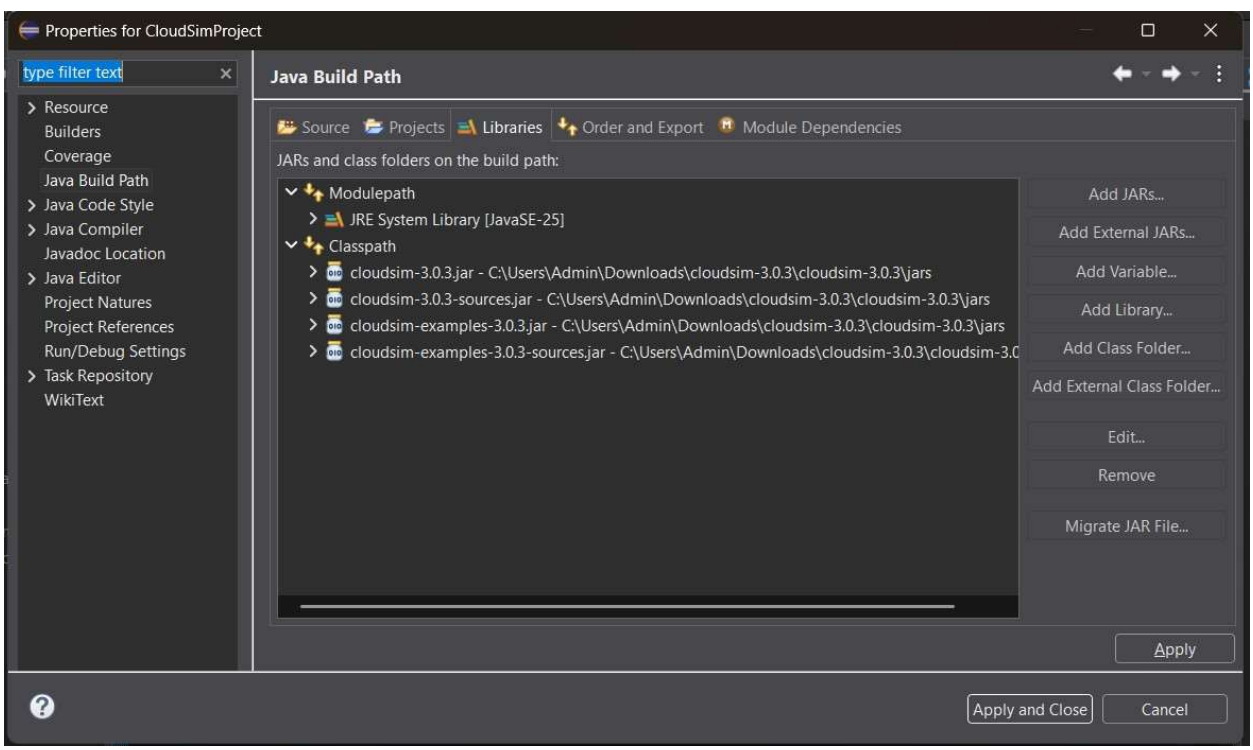

Fig 1.3: CloudSimProject properties

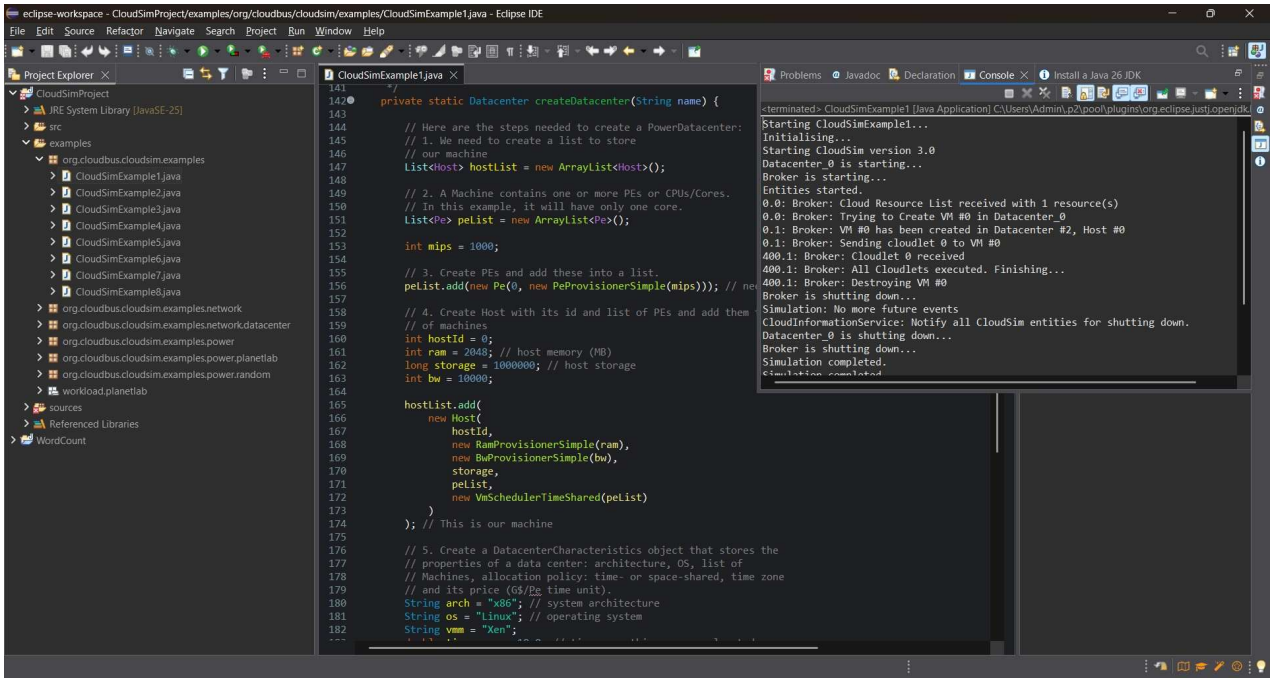

Fig 1.4: Running CloudSimProject - CloudSimExample1.java program

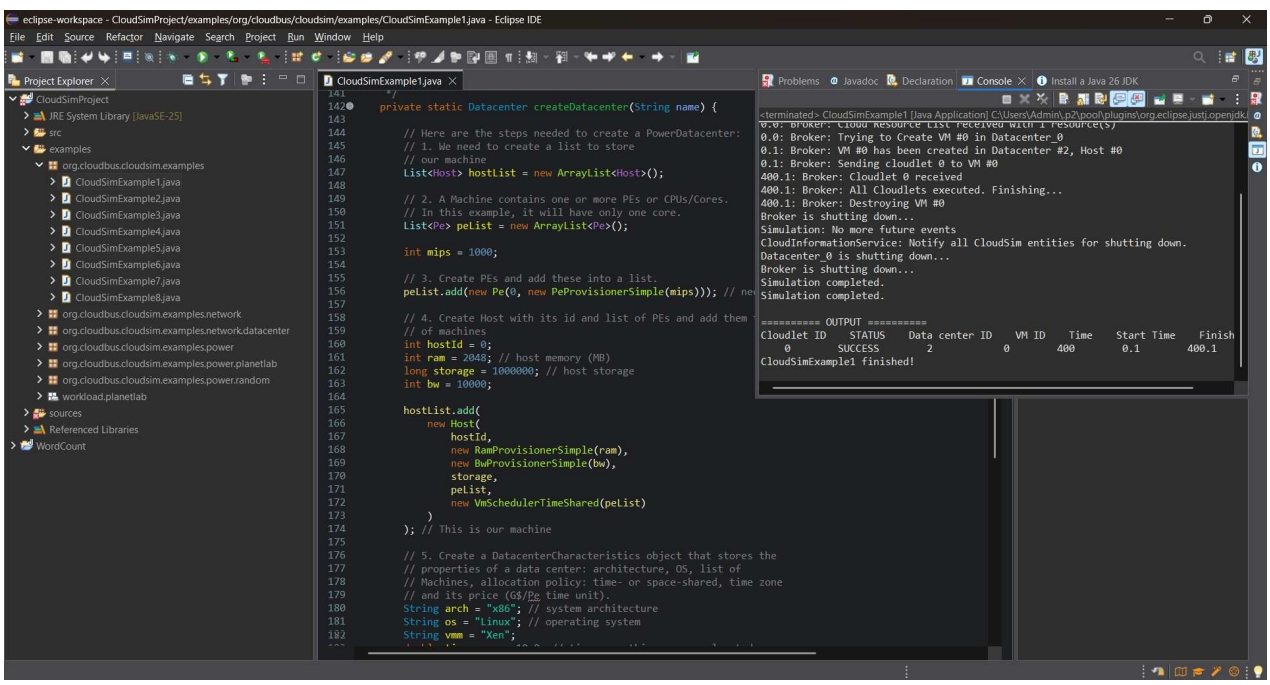

Fig 1.5: Simulation successful